

EX - 701
B.E. VII SEMESTER
EXAMINATION, DECEMBER 2014
POWER SYSTEM - II

TIME: THREE HOURS

MAXIMUM MARKS: 70

NOTE: ATTEMPT ALL QUESTIONS. ONE QUESTION FROM EACH UNIT. EACH QUESTION CARRY EQUAL MARKS.

UNIT - I

1. a) Explain the problems associated with modern interconnected power systems.
- b) Discuss the significance of distributed generation in a power system deregulation environment.

OR

2. Specify the following:
 - a) ATC
 - b) Congestion management
 - c) Deregulation

UNIT - II

3. a) Derive the necessary condition for economic operation of n-plants considering transmission losses.
- b) The incremental fuel costs of two units are

$$\frac{dC_1}{dP_1} = 0.1P_1 + 22; 20 \leq P_1 \leq 100MW;$$
$$\frac{dC_2}{dP_2} = 0.12P_2 + 16; 20 \leq P_2 \leq 100MW;$$

Determine the economic load allocation between the units when the total load on the system is 160 MW. C is in Rs/hr and P is in MW.

OR

4. Give reasons why
 - a) PV bus is considered as PQ bus at times in load flow solution process.
 - b) Real power cannot be transferred over lossless transmission line.
 - c) Load flow study requires slack bus.

UNIT - III

5. a) What is single area system? Deduce dynamic models for a single area system. Show the full block diagram.

b) Two generators rated 200 MW and 400 MW are operated in parallel. The drop characteristics of their generators are 4% and 5% respectively from no load to full load. Determine the load shared by each unit for a load of 500 MW.

OR

6. Specify the following

- a) Governor droop
- b) Load damping
- c) Tie line bias control
- d) ALFC
- e) ACE

UNIT - IV

7. a) What is an exciter. What is its role in AVR loop.

b) Explain AVR with the help of complete block diagram taking into account modern static excitation system of the alternator.

OR

8. a) Discuss in detail about generation and absorption of reactive power in power system components.

b) Explain reasons for variations of voltages in power system. Suggest any method for voltage profile improvement.

UNIT - V

9. a) Distinguish between rotor angle stability and voltage stability.

b) A 3-phase over head transmission line in which voltages at both ends are kept fixed at 400 kV has a negligible resistance and an inductive reactance of 20 ohms/ conductor. What is the maximum power that could be transmitted over this line. If this line when working at 60% of the above steady state limit suffers a short circuit which is cleared soon after and circuit restored. What are the limits of angular oscillations for transient stability?

OR

10. a) Derive Swing equation. Discuss its application.

b) Discuss the methods for improving transient stability.